Multi-Agent Reinforcement Learning for Renewable Energy Optimisation in Smart Cities

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Context & Problem

Why Smart Cities Need Smarter Energy Control

Cities now combine solar PV, distributed batteries, and intelligent grids — but coordinating them across buildings is hard. Mismatches between renewable supply and demand cause grid stress, wasted solar, and excess CO_2.

Three Core Challenges

- **Intermittent Generation:** Solar output fluctuates, creating unpredictable supply–demand gaps.
- **Siloed Control:** Buildings optimizing independently produce poor district-wide outcomes.
- **Peak Load Pressure:** Uncoordinated consumption spikes stress the grid.

Project Goals

- Deploy Multi-Agent Reinforcement Learning (MARL) for cooperative building-level control.
- Flatten district demand curves and maximize renewable self-consumption.
- Cut peak grid draw and CO_2 emissions across the full urban district.

The CityLearn Environment

CityLearn is a physics-based simulation platform for district energy management, built on real-world building datasets. Each building has solar PV, battery storage, thermal storage, and time-varying demand — all sharing a common grid.

The task: train one **smart controller per building** to optimally charge and discharge storage given prices, solar availability, and district load. Reinforcement Learning is ideal here — agents learn adaptive strategies from reward feedback, without hand-crafted rules, and generalize across seasons.

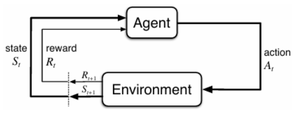


Figure 1: RL Paradigm

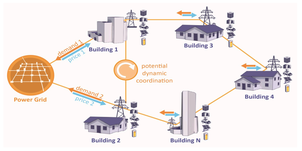


Figure 2: CityLearn Environment

Action Space — Each agent controls **3 continuous actuators** per timestep ($[-1, +1]$: charge → discharge):

Action	Effect
Cooling Storage	Shift cooling load via thermal tank charge/discharge
DHW Storage	Shift hot water load via tank charge/discharge
Electrical Battery	Store or release energy from PV or grid

Reward Functions — Three formulations, representing increasing coordination:

Reward	Description
Indepandent SAC	Each building minimizes **its own** net consumption. **No coordination** — pure selfish baseline.
SolarPenalty	Adds a **penalty for exporting excess PV**, pushing agents toward local solar self-consumption.
MARL	**District-level** reward penalizing net load variance across all buildings — explicit coordination incentive.

Key Performance Indicators (KPIs)

KPI	Target
Electricity Cost	Minimize total district expenditure
Peak Grid Demand	Reduce instantaneous power spikes
Renewable Utilization	Maximize solar self-consumption
CO_2 Emissions	Lower grid-sourced carbon footprint

Experiments

Three MARL strategies are benchmarked on a 3-building district, with the best then scaled to 6 buildings. All use official **CityLearn 2023 Challenge** environments built on real-world data:

- `phase_2_local_evaluation` — 3-building district (algorithm development & reward ablation)
- `phase_3_3` — 6-building district (scalability evaluation)

Algorithms at a Glance

- **Independent SAC** — One policy per building; local observations only; no information sharing. **Selfish baseline.**
- **Shared Policy SAC** — All buildings share **one policy**, pooling experience across buildings while still acting independently at runtime. Tested across all three reward functions to isolate reward design effects.
- **MARLISA** — Extends the shared policy with two **lightweight coordination signals** added to each agent's observation: (1) normalized net demand of all other buildings, and (2) a dispatch-order coefficient. Enables **implicit coordination without a centralized controller**.

Section 1 — Independent SAC Reward

Both agents optimized for their own net consumption only — no cooperative signal.

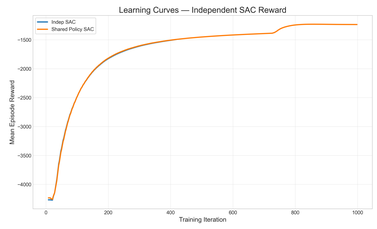


Figure 3: Learning curves — Indep SAC Reward

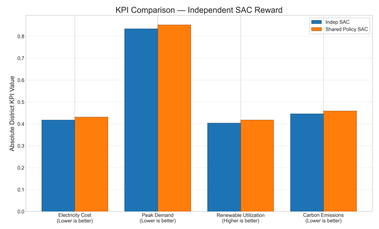


Figure 4: KPI comparison — Indep SAC Reward

Key finding: Shared Policy SAC and Indep SAC converge at the same rate and reach comparable KPIs. **Policy sharing alone, without a cooperative reward, yields no district-level benefit.** → Motivates the next reward designs.

Section 2 — Solar Penalty Reward

An export penalty pushes agents toward local solar self-consumption. MARLISA is introduced here.

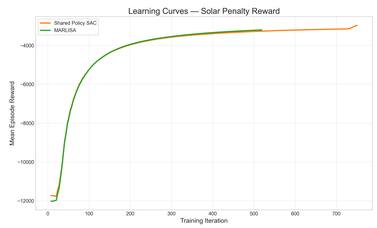


Figure 5: Learning curves — Solar Penalty Reward

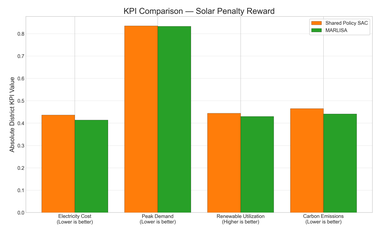


Figure 6: KPI comparison — Solar Penalty Reward

Key finding: MARLISA and Shared SAC follow near-identical **training** curves — yet MARLISA scores better on electricity cost, carbon, and renewable utilization at evaluation. **Coordination signals improve decision quality, not just training speed.**

Experiments (cont.)

Section 3 — MARL Reward

A district-level variance penalty directly rewards collective load flattening.

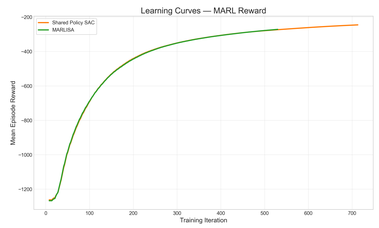


Figure 7: Learning curves — MARL Reward

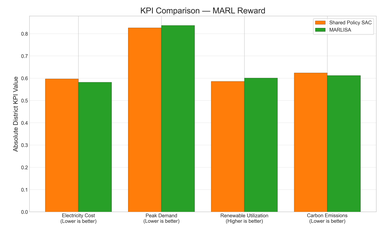


Figure 8: KPI comparison — MARL Reward

Key finding: Training curves remain similar between both algorithms — confirming MARLISA's advantage is at **inference, not training**. Under the MARL reward, MARLISA achieves its strongest results: lowest cost, lowest emissions, highest renewable utilization. **Best configuration overall.**

Section 4 — Scalability: MARLISA on 6 Buildings

The winning setup (MARLISA + MARL reward) is applied to the 6-building benchmark with **zero architectural changes**.

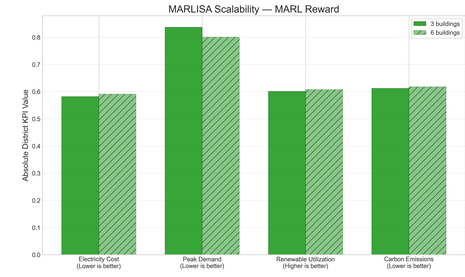


Figure 9: MARLISA — 3 vs 6 buildings, MARL Reward

Key finding: KPI performance holds as the district doubles in size. Peak demand is **further reduced** with more buildings — suggesting coordination signals grow more effective with agent count. **MARLISA scales gracefully.**

Conclusion

Lightweight coordination is enough. MARLISA — augmented with just two extra signals per agent — consistently outperforms both Independent and Shared Policy SAC across all rewards and KPIs, **without a centralized controller**.

Best Config	MARLISA + MARL Reward
Electricity Cost	↓ Lowest
Peak Demand	↓ Lowest
Renewable Use	↑ Highest
CO_2 Emissions	↓ Lowest

Gains scale to larger districts, confirming readiness for real smart city deployment.

Future work: larger benchmarks, solar forecasting integration, dynamic district topologies.

References

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Code : https://github.com/BrouthenKamel/huawei-degourdi